

The Bill Belongs to the Fan, Not the Path

A Conservation Law for Counterfactual Cost in Causal Zero Repair

Abstract

We isolate a simple conservation law hidden in exact causal zero repair. Let $\mathcal{B}_r$ be the minimum expected reserve required for r future steps, and let B_h be the reserve law after a history h . The excess

$$\Delta_h := \mathbb{E}B_h - \mathcal{B}_{r(h)}$$

above the optimal frontier is nonnegative. For every node with r steps remaining and every $s \leq r$ we prove

$$\sum_{|g|=s} P(g) \Delta_{hg} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-s}.$$

Thus the exponentially many possible continuations do not carry independent costs: they decompose one scalar frontier defect. If one continuation is itself optimal for the shorter horizon, then it carries zero excess and the entire marginal cost is borne by the counterfactual branches. In this precise sense, the causal bill belongs to the fan of possible futures, not to the realised path.

1 Setup

Let P and Q be probability laws of finite mean, and assume

$$\mathbb{E}P = \mathbb{E}Q. \tag{1}$$

The motivating example is the quartic pair

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}\right), \quad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64}\right),$$

for which

$$G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}.$$

The result below, however, uses only (1).

An exact depth- n zero-repair tree is a family of probability laws

$$\{B_h\}_{|h| \leq n}$$

such that at every internal history h ,

$$Q * B_h = \sum_x P(x) \delta_x * B_{hx}. \tag{2}$$

Let $\mathcal{K}_r$ denote the set of possible roots of exact depth- r trees, and define the optimal bill

$$\mathcal{B}_r := \inf_{B \in \mathcal{K}_r} \mathbb{E}B, \quad \mathcal{B}_0 := 0.$$

For a node h in a depth- n tree let

$$r(h) := n - |h|$$

be its remaining horizon. Since the subtree below h is itself feasible for $r(h)$ steps,

$$B_h \in \mathcal{K}_{r(h)},$$

and therefore

$$\mathbb{E}B_h \geq \mathcal{B}_{r(h)}.$$

This suggests the quantity

$$\boxed{\Delta_h := \mathbb{E}B_h - \mathcal{B}_{r(h)} \geq 0}, \tag{3}$$

which we call the *optimal-frontier defect*.

For a word $g = (g_1, \dots, g_s)$ write

$$P(g) := \prod_{j=1}^s P(g_j).$$

2 The conservation law

Theorem 1 (Conservation of the optimal-frontier defect). *Let $\{B_h\}_{|h|\leq n}$ be any exact depth- n zero-repair tree. If a node h has r steps remaining, then for every $0 \leq s \leq r$,*

$$\boxed{\sum_{|g|=s} P(g) \Delta_{hg} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-s}.} \quad (4)$$

In particular,

$$\boxed{\sum_x P(x) \Delta_{hx} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-1}.} \quad (5)$$

Proof. Take expectations in the exact recursion (2). Since convolution adds means,

$$\mathbb{E}Q + \mathbb{E}B_h = \sum_x P(x)(x + \mathbb{E}B_{hx}).$$

Hence

$$\mathbb{E}Q + \mathbb{E}B_h = \mathbb{E}P + \sum_x P(x) \mathbb{E}B_{hx}.$$

By (1),

$$\boxed{\mathbb{E}B_h = \sum_x P(x) \mathbb{E}B_{hx}.} \quad (6)$$

Iterating (6) for s generations gives

$$\mathbb{E}B_h = \sum_{|g|=s} P(g) \mathbb{E}B_{hg}. \quad (7)$$

Every descendant hg has remaining horizon $r - s$, so by (3),

$$\mathbb{E}B_{hg} = \mathcal{B}_{r-s} + \Delta_{hg}.$$

Substitution into (7) yields

$$\mathbb{E}B_h = \sum_{|g|=s} P(g) (\mathcal{B}_{r-s} + \Delta_{hg}).$$

Since

$$\sum_{|g|=s} P(g) = 1,$$

we obtain

$$\mathbb{E}B_h = \mathcal{B}_{r-s} + \sum_{|g|=s} P(g) \Delta_{hg}.$$

But also

$$\mathbb{E}B_h = \mathcal{B}_r + \Delta_h.$$

Equating the two expressions gives

$$\sum_{|g|=s} P(g) \Delta_{hg} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-s},$$

which is (4). Taking $s = 1$ gives (5). $\square$

The theorem compresses an entire future fan into one scalar identity. There may be exponentially many continuations, but their frontier defects cannot vary independently.

In particular, no additional collection of academic titles is required to keep track of them.¹

¹A minor tribute to the satirical Neczyklopedie portrait of Henryk Lahola, whose titulature is considerably less economical than (4).

3 The bill is counterfactual

Call h a *frontier contact* if

$$\Delta_h = 0.$$

Equivalently, the subtree rooted at h achieves the optimal bill for its remaining horizon.

The equality case of Theorem 1 is the central consequence.

Corollary 2 (Counterfactual bill). *Suppose that h is a frontier contact with r steps remaining and that, for some word g_* of length s , the descendant hg_* is again a frontier contact. Then*

$$\boxed{\mathcal{B}_r - \mathcal{B}_{r-s} = \sum_{\substack{|g|=s \\ g \neq g_*}} P(g) \Delta_{hg}.} \quad (8)$$

Proof. Since $\Delta_h = 0$, Theorem 1 gives

$$\mathcal{B}_r - \mathcal{B}_{r-s} = \sum_{|g|=s} P(g) \Delta_{hg}.$$

Since hg_* is also a frontier contact,

$$\Delta_{hg_*} = 0.$$

Its term therefore vanishes, leaving exactly (8). □

Equation (8) has a direct interpretation.

The continuation g_* which remains optimal carries *no excess cost*. Nevertheless the longer horizon is more expensive by

$$\mathcal{B}_r - \mathcal{B}_{r-s}.$$

That entire difference is carried by the other continuations which had to remain feasible even though they were not realised.

Hence

$$\boxed{\text{the bill is a property of the fan, not of the path.}}$$

This is not a metaphor added after the calculation. It is precisely the content of (8).

For one step, writing

$$d_r := \mathcal{B}_r - \mathcal{B}_{r-1},$$

a contact child hx_* gives the particularly transparent identity

$$\boxed{d_r = \sum_{x \neq x_*} P(x) \Delta_{hx}.} \quad (9)$$

Thus the marginal price of one additional future step is exported entirely into the sibling futures whenever the selected continuation stays on the optimal frontier.

4 A martingale form

Let

$$H_t = (X_1, \dots, X_t), \quad X_i \stackrel{\text{iid}}{\sim} P,$$

be a random branch of an optimal depth- n tree.

Corollary 3 (Renormalised defect martingale). *The process*

$$\boxed{Z_t = \Delta_{H_t} - (\mathcal{B}_n - \mathcal{B}_{n-t})}$$

is a martingale. Consequently,

$$\boxed{\mathbb{E} \Delta_{H_t} = \mathcal{B}_n - \mathcal{B}_{n-t}.} \quad (10)$$

Proof. Conditioning on H_t and applying (5),

$$\mathbb{E}[\Delta_{H_{t+1}} \mid H_t] = \Delta_{H_t} + \mathcal{B}_{n-t} - \mathcal{B}_{n-t-1}.$$

Subtracting

$$\mathcal{B}_n - \mathcal{B}_{n-t-1}$$

from both sides gives

$$\mathbb{E}[Z_{t+1} \mid H_t] = Z_t.$$

At an optimal root $\Delta_\emptyset = 0$, hence $Z_0 = 0$ and (10) follows. $\square$

5 Conclusion

The conservation law separates two notions which are easily conflated: the history which is eventually realised and the family of histories which must remain simultaneously possible.

An exact causal mechanism pays for the latter.

At a frontier contact the realised continuation may itself remain perfectly optimal, carrying zero frontier defect, while the entire incremental cost of the longer horizon is stored in its unrealised siblings.

The exponentially branching tree therefore does not represent an exponential collection of independent bills. Its branches are a decomposition of one conserved counterfactual cost.

The bill belongs to the fan, not the path.